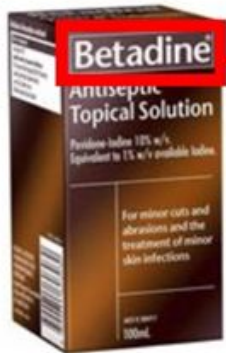




BODREX
MIGRA



BETADINE



AMOXSAN

Images-based Drug Search Engine using OCR and Levenshtein Algorithm Spelling Correction



BODREX
EXTRA

Problem Background

In healthcare, the multitude of drug brands poses a challenge for professionals who must swiftly and **accurately identify them**. This is crucial for administering the right medication, preventing adverse reactions, and ensuring optimal patient outcomes. The diversity of drugs makes **quick identification essential** for pharmacists, nurses, and physicians.



To address this, **Optical Character Recognition (OCR) technology** is increasingly employed, automating the process and aligning with the broader trend of digitization in healthcare.

Dataset

Train Dataset



Test Dataset



OCR Model

We're using **PaddleOCR** as the pretrained model

Problem:

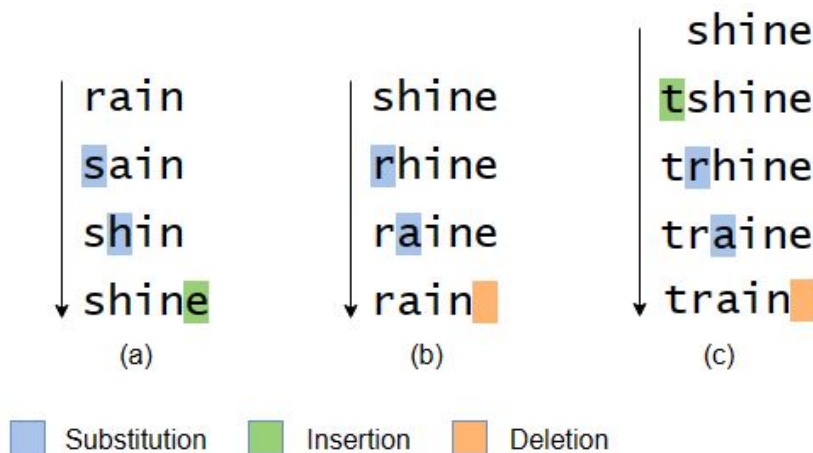
- There are **some words** on the brand that **can't be detected**
- **Detects all word** on the packaging



1: AMBER 0.982
2: M 0.955
3: PHARMACEUTICALS, INC. 0.982
4: NDC 31722-562-24 0.999
5: Abacavir Oral 0.985
6: Solution USP 0.987
7: 20 mg/mL* 1.000
8: Notice to Authorized Dispenser:. 0.980
9: Each time Abacavir Oral Solution is dispensed, 0.996
10: give the patient a Medication Guide and 0.996
11: Warning Card from the carton. 0.968
12: Rx Only 0.918
13: 240 mL 0.924

Solution? Spelling Correction

Utilize **Levenshtein Distance Algorithm** to find the closest brand from the **database** (all brands from training dataset) with the detected words



Brand Database

Collect all the brands from train dataset to create a database

Brands
ABACAVIR
ACARBOSE
ACETYLCYSTEINE
...

Modeling & Results

Metrics	Score
Accuracy	82.57%
WER	0.19475
CER	0.16757

Based on the results, there are still some **error**. We can improve the result by doing **post-evaluation** on the dataset and prediction.

Post Evaluation: Inconsistency in brand names (1)

Example



Glucobay



Has the same meaning as **Acarbose**

Else	Precose	→	Has the same meaning as Acarbose
	Humira	→	Has the same meaning as Adalimumab
	Mallinckrodt	→	Has the same meaning as Codipront
	Ensidol	→	Has the same meaning as Etodolac
	Anafranil	→	Has the same meaning as Etodolac

There are some drug images that **don't have the brand name that should be predicted**, but it **has the same meaning as the brand that should be predicted**.

Post Evaluation: Inconsistency in brand names (2)



- 1: 32 0.998
- 2: PARAWOL 0.963
- 3: TABI 0.978
- 4: IHYDROCODEINE 0.992
- 5: PARACETAMOI 0.961
- 6: YOU CAN'T HIT PAIN 0.897
- 7: CASYTO 0.627
- 8: MUCH HARDER 0.965

The expected label is “**Paramol Forte**” but the given image only has “**Paramol**” text on it. So we have to handle it!

Post Evaluation: Tricky label (1)



1: Calasine 0.970
2: Calamine 0.996
3: lotion 0.995
4: ergpadakult 0.865

- It can be seen in the image below, that the word "**Caladine**" (which is wrongly predicted by the model to be **Calasine**) is separated from the word "**Lotion**", so this can cause the edit distance of "**Caladine**" to "**Caladine Lotion**" to be **further away**.
- Whereas in the word "**Calamine**", there is also the word "**Calamine**" in the database so the edit distance becomes closer.
- To solve this problem, the word "**Calamine**" can be **removed from the detected text**, so that the closest word is the correct one.

Post Evaluation: Tricky label (2)

```
1: Krim 0.677
2: fungiderm 0.997
3: neto:10g 0.802
4: Clotrimazole 0.996
5: UNTUK INFEKSIJAMUR 0.933
6: KONIMEE 0.888
```



- On the drug “Fungiderm”, it says “Clotrimazole”. The problem is that “Clotrimazole” is a drug that is on our database so spelling correction can be off.
- The solution is to remove “Clotrimazole” from the detected text

Post Evaluation: Irrelevant Images



1: akar mata ikan yang 0.975
2: sudah mati akan 0.973
3: menghitammengering 0.993
4: dan mengelupas... 0.927

- It can be seen in the testing dataset that the testing data sometimes contains images that are **not relevant to the text** (in this case OCR), so it will be an **impossible task** to do. From there our team found that the testing dataset has data that has **the same label sequentially every 10 data**, so we can do **post evaluation** by obtaining the **mode of each of the 10 data** and replacing the data with the mode.
- For example, as in the image on the left, where the image has no text relevant to the label. However, we as humans understand that this image is still connected to the label around it, namely **"Callusol"** which is the medicine for the above disease.

Modeling & Results After Post Evaluation

Metrics	Score
Accuracy	100%
WER	0.0
CER	0.0

After conducting a post-evaluation, we successfully **rectified all errors** in the dataset, resulting in a **flawless assessment** of our approach.

A person in a dark suit is seen from the side, writing on a wall. The wall is covered with numerous yellow sticky notes, each containing handwritten text. The person's hand is holding a black marker. The scene is overlaid with a semi-transparent blue filter. The text 'Thank You' is prominently displayed in the center in a large, white, sans-serif font.

Thank You

CONCLUDING
THE FELLOWSHIP

FOLLOW-UP
AFTER THE
FELLOWSHIP IS
OVER

FUTURE
OPPORTUNITIES

I NEVER
KNEW IT
WAS OVER
NO COMPLETION
DATE BETTY

THE NETWORK
WAS GREAT.
BEING IN NEW YORK
CITY.

THE EXPOSE
ADDS VALUE
TO MY
BRAND

Flexibility
of Proj
was (beneficial)

RUNNING
POINTS

REPORTS
RECOUNTING

ATTENDANCE
REPORTS

THE REPORTS
WAS MORE
THAN
ACADEMIC

SPENT MORE
TIME
BEING CREATIVE

THANKS
FOR
MARKETING

ALL THE
INFORMATION
AT THE END
WAS HARD
MISTAKES
FRAMING

FROM OTHERS
TO "GIVEN UP" IF
REPORT

TICKETING
PLAN FOR
PEOPLE WHO
CAME

I WOULD
HAVE MADE
IT PRACTICAL
IF IT WAS
PRACTICAL

I DON'T
WANT TO
SOUND LIKE
A PRO

PROFESSIONAL
STRUCTURE